

Review

Assessing executive functioning in individuals with social anxiety disorder (SAD) across the lifespan: A systematic literature review and meta-analysis

Alexandra A. Harrison ^{*} , Lan Nguyen, Karen Murphy, David Neumann

School of Applied Psychology, Griffith University, Gold Coast, QLD 4222, Australia

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ABSTRACT

Social anxiety disorder is a pervasive clinical disorder characterised by intense fear and/or avoidance of one or more social situations, and has been linked to deficits in executive functioning performance. However, methodological differences and mixed results have made it difficult to draw definitive conclusions from individual studies. The current systematic review and meta-analysis collated the results from 49 studies to examine the link between social anxiety disorder and executive functioning across the lifespan. Findings revealed that individuals with social anxiety performed significantly worse than healthy controls or low social anxiety groups on measures of executive functioning ($r = -.15$), specifically cognitive flexibility ($r = -.20$), inhibitory control ($r = -.18$), and global executive functioning ($r = -.17$). No significant association was evident between working memory and social anxiety ($r = -.06$). In addition, the type of measure (self-report vs cognitive task) moderated the relationship between social anxiety and executive functioning. Although age did not moderate the overall relationship, the association between the individual domains of executive functioning and social anxiety differed between youth and adults, which may reflect the different developmental timelines between the domains across the lifespan. The findings offer valuable insight into our understanding of the development of executive functions for individuals with social anxiety and could assist with forming new strategies or interventions to improve daily functioning in this clinical population.

Social anxiety disorder (SAD) is a clinical disorder characterised by marked fear and/or avoidance of one or more social situations, in which an individual feels scrutinised or judged by others (Stein & Stein, 2008). Specifically, individuals with SAD fear acting in a certain way, or displaying physical manifestations of anxiety, that could lead to embarrassment or social rejection (Hedges et al., 2019; Morrison & Heimberg, 2013). Research indicates that SAD is one of the most widespread mental health disorders, with 12-month and lifetime prevalence rates of 7.1 % and 12.1 %, respectively (Kessler et al., 2005). The pervasiveness of SAD is of concern given that this disorder is associated with clinically significant distress and impaired functioning in social, occupational, and cognitive domains (Aderka et al., 2012; American Psychiatric Association, 2013), likely stemming from deficits in executive functioning (Demetriou et al., 2018; Fujii et al., 2013).

Executive functions (EFs) are considered to be the most complex form of high-level human cognition and include a group of cognitive skills that facilitate the completion of goal-directed behaviour (Diamond, 2016; Friedman & Miyake, 2017; Miyake et al., 2000a).

Research suggests that there are three core EFs: working memory, cognitive flexibility, and inhibitory control (Miyake et al., 2000b). Working memory refers to the ability to temporarily store information for immediate mental use, cognitive flexibility refers to the ability to switch between and adapt to changes in task demands, and inhibitory control refers to the ability to regulate one's thoughts and/or behaviours in response to stimuli and the goals of a task (Diamond, 2013; Miyake et al., 2000b). EFs follow distinct developmental trajectories across the lifespan, with EF performance differing between children, adolescents, adults, and older adults (Korzeniowski et al., 2021; Petrican et al., 2017). This development corresponds with the maturation of key brain regions in the frontal lobes needed to support EFs (Conklin et al., 2007; Magar et al., 2010). The continued development of EFs and their associated brain regions occur throughout adolescence and early adulthood, at which point peak maturation is reached, which is followed by a plateau in performance and a subsequent decrease in functioning later in life (Huizinga et al., 2006). In addition, there are distinct developmental trajectories for each of the three core EFs, which are thought to reach

^{*} Corresponding author.

E-mail addresses: Allie.Harrison@griffithuni.edu.au (A.A. Harrison), Lan.Nguyen@griffith.edu.au (L. Nguyen), K.Murphy@griffith.edu.au (K. Murphy), D.Neumann@griffith.edu.au (D. Neumann).

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maturity between preadolescence and early adulthood (Boelema et al., 2014; Huizinga et al., 2006). Specifically, working memory continues to develop until the age of 25 (Gur et al., 2012; Luna et al., 2004), cognitive flexibility until the age of 17 (Boelema et al., 2014; Magar et al., 2010) and inhibition until the age of 15 (Huizinga et al., 2006; Luna et al., 2004) in healthy individuals.

According to neurological and cognitive research, the development of EFs across the lifespan may differ for individuals with SAD compared to typically developing individuals (Haller et al., 2015; Kim et al., 2011; Schmidt et al., 2022). Studies show that adolescents with SAD exhibit reduced brain maturation in the frontal brain regions, which may result in either a delay in the development of EFs or a premature plateau, preventing them from reaching peak EF performance (Haller et al., 2015; Schmidt et al., 2022). The differing brain maturation could explain the observed cognitive deficits for individuals with SAD (Ding et al., 2011; Kim et al., 2011). As a result, age group differences in EF performance may be evident for individuals with SAD compared to healthy controls (HCs), and therefore, differences in EF across the lifespan should be explored.

To date, studies that have examined the link between SAD and EF have reported mixed findings. Some studies have shown that adults with SAD perform significantly worse than HCs on measures assessing working memory (Abushalmaq et al., 2021; Asmundson et al., 1994; Bourke et al., 2012; Demetriou et al., 2018), cognitive flexibility (Abasi et al., 2017; Cohen et al., 1996; Du et al., 2022; Fujii et al., 2013; Topcuoglu et al., 2009), and inhibitory control (Cao et al., 2022; Demetriou et al., 2018; Rosa-Alcázar et al., 2021). In contrast, other researchers have reported that adults with SAD do not perform significantly differently from HCs on these EF measures (Asmundson et al., 1994; Bourke et al., 2012; Graver & White, 2007; Grisham & Williams, 2013; Judah et al., 2016; O'Toole & Pedersen, 2011; Whitsitt, 2022), or even perform significantly better (Demetriou et al., 2021; Dotson et al., 2014; Jiang et al., 2022; Moriya, 2018; Moriya & Sugiura, 2013). Studies with children and adolescents have reported similar inconsistencies (Castagna et al., 2019; Gómez-Ariza et al., 2013; Morea & Calvete, 2021, 2023), although there is limited research in these age groups.

The inconsistency in findings regarding the association between SAD and EFs may reflect the influence of other variables that moderate the relationship. As discussed, each core EF is shown to follow a distinct developmental trajectory across the lifespan which may differ for individuals with social anxiety. As a result, any heterogeneity in effect sizes across the studies may be explained by variability in the ages of participants. In addition, SAD severity, specifically the use of clinical versus non-clinical samples, may impact findings. For example, clinical samples may produce stronger associations between SAD and EF as they typically present with more varied and extreme symptoms, while non-clinical samples are more homogenous and are at the lower end of the spectrum of symptomatology. Furthermore, comorbidity may also be a relevant moderator, with research suggesting that comorbid clinical disorders can further increase cognitive impairments, with these individuals performing worse on measures of EF than individuals presenting with social anxiety alone (Graver & White, 2007; Musa et al., 2003).

The use of self-report as opposed to behavioural measures may be another moderating factor. SAD is commonly associated with maladaptive cognitions that have been shown to have detrimental effects on individuals' perception of their own functioning (Ferreri et al., 2011; Kuru et al., 2018). This negative perception of self may extend to their subjective cognitive evaluations, reflecting poorer self-reported EF performance (Allott et al., 2020; Demetriou et al., 2018). As a result, these individuals may report greater deficits than indicated by their cognitive performance, and therefore, the influence of assessment type must be considered.

Finally, task index (i.e., the outcome measure used to evaluate participants performance on a specific task, such as accuracy or reaction

time) may be important for comparisons between individuals with SAD and HCs. Attentional control theory (ACT) posits that rather than impairing performance effectiveness (the quality of performance as indicated by response accuracy measures), anxiety is associated with deficits in processing efficiency (the ratio of response accuracy to the use of cognitive resources required to complete a task) (Derakshan & Eysenck, 2009; Eysenck & Derakshan, 2011; Eysenck et al., 2007). This is because highly anxious individuals often use alternative strategies, such as enhanced effort or greater use of processing resources, to compensate for anxiety-induced cognitive deficits (Derakshan & Eysenck, 2009). This leads to normal accuracy but slower response times during EF tasks (Eysenck & Derakshan, 2011; Eysenck et al., 2007). As a result, studies that only measure one task index may produce biased findings when comparing SAD and HCs. Taken together, task index, comorbidities, assessment type, diagnostic status of participants, and participant age are all potential moderators that should be considered and accounted for to provide a more comprehensive examination of the link between SAD and EF.

To date, four systematic reviews and meta-analyses have examined the link between SAD and EF. Shi et al. (2019) conducted a meta-analysis of 58 studies to examine the relationship between anxiety and attentional control, and found that significant deficits were evident in inhibitory control and cognitive flexibility, but not working memory, for individuals with SAD. Similarly, Dorenkamp and Vik (2018), conducted a systematic review of 78 studies that examined the link between anxiety and neuropsychological test performance, revealing that anxiety was associated with poorer verbal memory, attention, inhibition and working memory. In addition, O'Toole and Pedersen (2011) conducted a systematic review of neuropsychological performance in SAD. Their review of 30 articles revealed that SAD was significantly associated with poorer EF performance. Finally, Majeed et al. (2023) recently published a three-level meta-analysis that examined the link between anxiety disorders and EFs across 55 studies. This analysis found that individuals with anxiety reported significantly poorer performance efficiency on EF tasks, while performance effectiveness remained intact, which is consistent with the ACT. However, when examining SAD specifically, there was no significant difference in EF performance compared to HCs.

While past reviews have been conducted to examine the relationship between SAD and EF in adult populations, research across the lifespan has been overlooked. The lack of focus on children and adolescents is of concern given that these developmental stages represent the typical age of onset for SAD symptoms and EF development (Elizabeth et al., 2006; Theodoraki et al., 2020; Vogel et al., 2023). In addition, these reviews did not examine individual cognitive domains of EF in relation to social anxiety (Dorenkamp & Vik, 2018; Majeed et al., 2023; O'Toole & Pedersen, 2011; Shi et al., 2019), which is more comprehensive and could offer a more nuanced understanding of the relationship between SAD and EF. Moreover, task index was only examined in relation to overall EF (Majeed et al., 2023) and no reviews have examined potential differences between self-report and cognitive tasks. Such factors are important and may explain the mixed findings reported in the literature to date.

Therefore, the current systematic review and meta-analysis aimed to examine the link between SAD and EF by synthesising the existing research and providing an integrated summary of the EF of this clinical group across the lifespan. This meta-analysis also aimed to explore the potential influence of moderator variables, specifically age, SAD severity, assessment type (self-report vs. cognitive task), comorbidity, and task index (reaction time vs. accuracy) on this relationship. This will allow for an overview of any differences in EF development, and relevant gaps and limitations to be identified within the literature. The findings could also inform future research on the psychopathology of SAD and could assist with the development of new strategies to optimise treatment outcomes and improve both cognitive and overall functioning in this clinical population.

1. Method

1.1. Eligibility criteria

The following criteria were used to guide the inclusion or exclusion of studies for the review and were developed in accordance with PRISMA guidelines (Page et al., 2021) and existing research (Diamond, 2013; Majeed et al., 2023; Stein & Stein, 2008). Empirical studies that examined the relationship between SAD and EF using a correlational (cross-sectional or longitudinal) or experimental/quasi-experimental design (pre-test – intervention – post-test) where assessments of social anxiety and EF were completed at pre-test, were considered for inclusion. Studies must have included a SAD group, as indicated by a clinical diagnosis or established measure of SAD (e.g., high levels of social anxiety according to the Liebowitz Social Anxiety Scale), and a healthy control or low social anxiety group for comparison, or examined SAD symptoms on a continuum. Participants of interest included children (8–12 years), adolescents (13–19 years), adults (20–64 years), and older adults (65 years and above (Petrican et al., 2017). In addition, the studies must have included a behavioural or self-report measure of at least one EF domain (i.e., global or one of the three core EFs: working memory, cognitive flexibility, and inhibition; (Diamond, 2013; Miyake et al., 2000b), and be published in English. Studies that did not analyse SAD separately (i.e., combined sample with other anxiety disorders), or that included SAD participants with any other neurological, psychiatric, or medical issues, were excluded. Only peer reviewed sources were considered for inclusion in this review.

2. Search method and study selection

Searches were conducted in Scopus, PubMed (Ovid), PsycINFO (Ovid), ProQuest Health and Medical, and ProQuest Psychology. The following search terms were entered into each database to identify relevant studies: (“youth” OR “young person*” OR “teen” OR “early adulthood” OR “child*” OR “adolescen*” OR “preadolescen*” OR “lifespan” OR “old* adult*” OR “adult*” OR “develop”) AND (“SAD” OR “social anxiet*” OR “social phobia” OR “social anxiety disorder”) AND (“executive function*” OR “EF” OR “executive performance” OR “central executive” OR “executive control” OR “cognitive performance” OR “neuropsychological function” OR “neuropsychological test” OR “executive dysfunction” OR “cognitive flexibility” OR “shifting” OR “task switch*” OR “mental flexibility” OR “working memory” OR “WMC” OR “updating” OR “inhibition” OR “inhibitory control”). Subject headings were also used in the search for relevant databases (see Appendix A). Database searching was conducted on June 25th, 2024 and updated on June 18th, 2025.

Additional search strategies were employed including ascendancy and descendancy approaches conducted through reference lists of relevant studies. This was completed once prior to the final search and once after screening. Potential grey literature was also examined by conducting additional database searches in ProQuest Dissertations and Theses and ProQuest Conference Papers Index.

The screening process was completed by two independent reviewers using the Covidence Systematic Review Software (Veritas Health Innovation, n.d.). Studies were initially screened for eligibility by examining the titles and abstracts. Once potentially relevant records were identified, the full-text versions of these studies were documented and assessed for inclusion. There was strong inter-rater reliability between the reviewers (Cohen’s κ = 0.82). All discrepancies were resolved following discussion of the eligibility criteria.

3. Search results

The comprehensive search of the literature revealed a total of 4020 records (2482 following removal of duplicates). Of these studies, 58 were deemed eligible and included in the review. The authors of 14

studies were contacted for additional data as the original studies provided insufficient data for the analysis. Of the 14 authors, 5 responded and 9 did not provide additional information. These 9 studies did not differ in any meaningful way from the studies that were included (i.e., similar study design, methodology, participants, outcomes etc). Therefore, listwise deletion was used, and the bias introduced by their non-inclusion was limited. As a results, 49 studies were included in the final meta-analysis (see Fig. 1 for the PRISMA flow diagram and Appendix B for included studies).

4. Data analysis

4.1. Data extraction and management

Comprehensive Meta-Analysis (CMA) software was used to conduct all analyses (Borenstein et al., 2014). Data was extracted and entered into the software including means, standard deviations and sample sizes, or correlations (i.e., r -values). Some studies utilised several outcome measures to assess EF domains (i.e., Digit Span and Letter-Number task to assess working memory). In accordance with meta-analysis protocol, in these circumstances data were combined into a single effect size per study, which was used in the final analyses (Borenstein et al., 2009).

The overall effect size was calculated as Pearson’s correlation (r), with a positive effect size indicating that higher SAD symptoms/severity was associated with better EF performance, and a negative effect size indicating that higher SAD symptoms/severity was associated with poorer EF performance. According to the literature, an effect size of 0.1 indicates a small effect, 0.3 indicates a medium effect, and 0.5 indicates a large effect (Cohen, 1988).

4.2. Assessment of methodological quality

The methodological quality of each study was evaluated using the Mixed Methods Appraisal Tool (MMAT; (Hong et al., 2018). Studies were examined according to five criteria: i) sampling strategy, ii) the representativeness of the sample, iii) suitability of the measurements used, iv) risk of non-response bias, and v) appropriateness of statistical analyses. The MMAT provides a total score out of 5, with higher scores indicating better methodological quality. This evaluation process was completed by two independent reviewers, who reported 100 % agreement on the appraisal rating for the studies.

The Q -statistic was used to assess the heterogeneity of effect sizes. I^2 was used to represent the proportion of variance in the true effect size across the studies, with 0–40 %, 30–60 %, 50–90 % and 75–100 % representing low, moderate, substantial and high heterogeneity respectively (Deeks et al., 2024). Increased heterogeneity suggest that covariates may potentially explain a larger proportion of true variance (Borenstein et al., 2009). If significant heterogeneity was observed, moderator analyses were conducted to explore the potential sources of variance across effect sizes.

To examine publication bias, funnel plots were examined for asymmetry, and Egger’s test (Egger et al., 1997) was used to statistically support findings. Duval and Tweedie (2000) trim-and-fill statistics were also examined to further assess the potential influence of publication bias, providing an estimate of the unbiased effect size through removing outliers and imputing missing cases to generate a symmetrical funnel plot. It is suggested that if the adjusted effect size remains the same, the analysis is likely robust to the influence of publication bias. Alternatively, the impact of bias may be considered modest if the adjusted effect size changes, but the overall findings remain the same, and substantial if the key findings are altered as a result of the effect size adjusted for bias (Borenstein et al., 2009). The original and adjusted effect sizes were reported in the current meta-analysis.

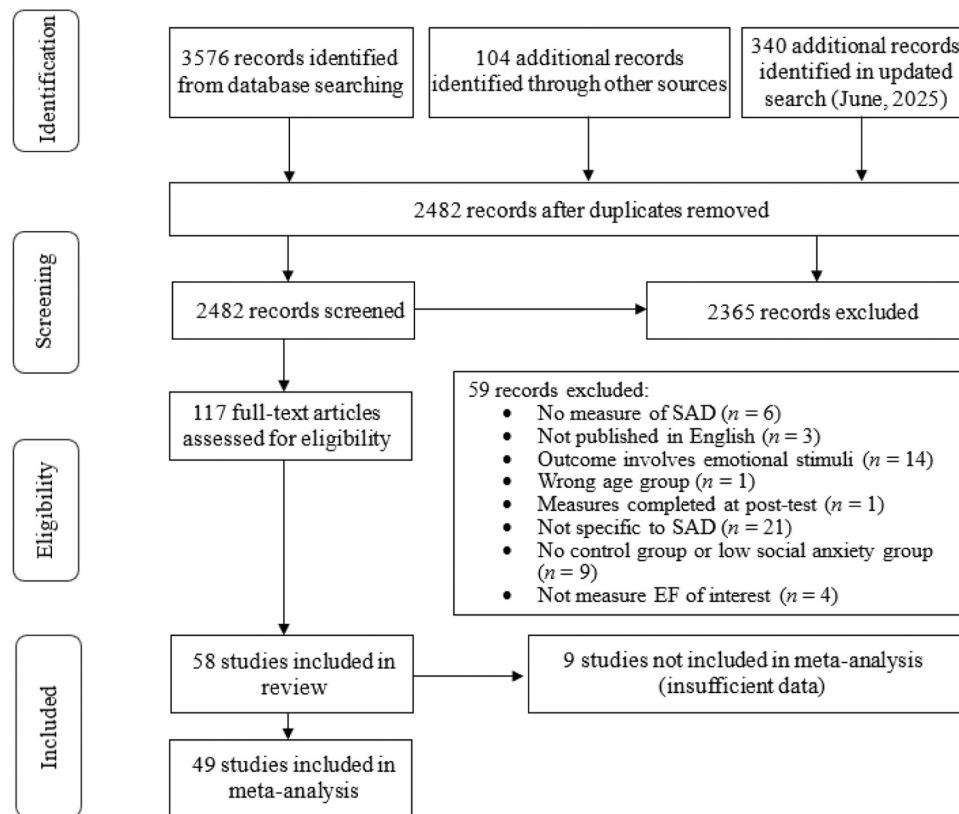


Fig. 1. PRISMA Flow Diagram Outlining the Study Screening and Selection Process.

4.3. Data synthesis

All analyses were performed based on the random-effects model due to variability in study design and outcomes, with sensitivity analysis indicating that the findings were robust (see Sensitivity Analyses section). An alpha level of .05 was used for all analyses. Separate analyses were conducted for the different EF domains (i.e., working memory, cognitive flexibility, inhibitory control, and global EF), to explore the link between social anxiety and specific EFs. Moderator analyses were also conducted for i) age (child/adolescent, adult), ii) SAD severity (clinical, non-clinical), iii) assessment type (self-report, cognitive task), iv) task index (reaction time, accuracy), and v) comorbidity (controlled or did not control for comorbidities).

5. Results

5.1. Characteristics of included studies

Forty-nine studies examined the link between social anxiety and EF in individuals at one or more points across the lifespan. There was a total of 5549 participants, with 43 studies focusing on adults ($N = 3654$, $M_{age} = 27.13$, $SD = 8.25$), and six studies focusing on children and adolescents ($N = 1895$, $M_{age} = 13.60$, $SD = 1.69$). The majority of studies were conducted in North America (34.69 %), followed by Asia (26.53 %), Europe (26.53 %), Australasia (10.20 %), and Africa (2.04 %). Twenty-one studies assessed EF outcomes in clinical populations, and 28 focused on non-clinical populations including low and high social anxiety groups. Of the 49 studies, 20 were correlational studies ($N = 3638$, $M_{age} = 19.93$, $SD = 6.72$) and 29 were experimental/quasi-experimental studies ($N = 1911$, $M_{age} = 28.05$, $SD = 8.52$) that used SAD groups ($N = 865$, $M_{age} = 28.20$, $SD = 8.49$) and HC comparison groups ($N = 1046$, $M_{age} = 27.90$, $SD = 8.71$).

All of the studies examined at least one of the three core EFs (working

memory, cognitive flexibility, or inhibitory control), or global EF, which were categorized in accordance with the Compendium of Neuropsychological Tests (Strauss et al., 2006). The most prominent domains measured throughout the literature were inhibitory control (38.78 %), followed by working memory (36.73 %), global EF (34.69 %), and cognitive flexibility (30.61 %). The majority of studies examined a single EF domain (67.35 %), and used a single measure to assess EF in their sample (67.35 %).

The studies included in this review exhibited mostly high methodological quality, with 20 studies obtaining MMAT scores of 4 and above (out of 5). The appraisal revealed 29 studies obtaining an MMAT rating of 3 (out of 5), which was typically due to recruitment bias or issues with potentially non-representative samples, and risk of non-response bias (see Table 1 for a summary of study characteristics).

6. Sensitivity analyses

Sensitivity analyses were performed to ensure that potentially influential statistical assumptions did not impact on the key findings of the meta-analysis. The overall mean effect size was small and significant for both the random ($r = -.15$, 95 % CI [-0.22, -0.08], $p < .001$), and fixed-effect models ($r = -.18$, 95 % CI [-0.21, -0.16], $p < .001$). As the interpretation remained the same, the findings were considered robust to model type. In addition, a sensitivity analysis was performed for methodological quality. The mean effect size remained small and significant for studies with both low ($r = -.18$, 95 % CI [-0.28, -0.07], $p = .001$) and high MMAT scores ($r = -.12$, 95 % CI [-0.22, -0.02], $p = .025$), and as a result, all studies were retained and used in the meta-analysis. A sensitivity analysis was also conducted to examine the impact of independence of errors, as recommended by researchers (Noble et al., 2017). The overall mean effect size was small and significant for both studies with multiple effect sizes ($r = -.13$, 95 % CI [-0.23, -0.03], $p = .010$), and those with single effect sizes ($r = -.20$, 95 % CI

Table 1
Summary of Study Characteristics and Findings for Studies Included in the Meta-Analysis.

Study	N	Age group	Country	Sample Characteristics	Measure of SAD	Population	EF of Interest	EF Measure	MMAT Score
Abasi et al. (2017)	524	Adults	Iran	524 adults ($M = 35.24$, $SD = 10.71$), 53.6 % males	Social Interaction Anxiety Scale (SIAS)	Non-clinical	Cognitive Flexibility	Attentional Control Scale (ACS)	4
Abushalbak et al. (2021)	44	Adults	Egypt	SAD group: 20 adults ($M = 45$, $SD = 4.24$), 55 % males HC group: 24 participants ($M = 43.92$, $SD = 6.83$), 58.33 % males	Mini International Neuropsychiatric Interview (MINI)	Clinical	Working Memory	Digit Span Forward and Backward, N-Back Task, Short-Delay Working Memory Task, Long-Delay Working Memory task	3
Airaksinen et al. (2005)	207	Adults	Sweden	SAD group: 32 adults ($M = 38$, $SD = 12.3$), 31.3 % males HC group: 175 adults ($M = 43.9$, $SD = 12.3$), 50.9 % males	Schedules for Clinical Assessments in Neuropsychiatry (SCAN)	Clinical	Cognitive Flexibility	Trail-Making Test - B (TMT)	4
Asmundson et al. (1994)	34	Adults	Canada	SAD group: 18 adults ($M = 38.4$, $SD = 9.2$), 72.22 % males HC group: 16 adults ($M = 34.9$, $SD = 11.0$), 56.25 % males	Modified Structured Clinical Interview for DSM-III-R diagnoses (SCID)	Clinical	Cognitive Flexibility, Working Memory	Trail-making test (TMT)	3
Bourke et al. (2012)	106	Adults	New Zealand	SAD group: 30 adults ($M = 37.1$, $SD = 13.1$), 63.33 % males HC group: 76 adults ($M = 38.1$, $SD = 13.0$), 34.21 % males	Structured Clinical Interview for DSM-IV diagnoses (SCID) Patient Edition	Clinical	Working Memory	Digit Span Forward and Backward, CANTAB Spatial Span, CANTAB Spatial Working Memory, COWAT Sum of Words Stroop task	4
Cao et al. (2022)	80	Adults	China	80 adults ($M = 20.7$, $SD = 2.0$), 17.5 % males	Liebowitz Social Anxiety Scale (LSAS)	Non-clinical	Inhibitory Control		4
Castagna et al. (2019)	159	Children/Adolescents	USA	159 children/adolescents ($M = 10.97$, $SD = 2.38$), 60.38 % males	Multidimensional Anxiety Scale for Children (MASC)	Non-clinical	Working Memory	Digit Span Backward, Letter-Number Sequencing	3
Claus et al. (2023)	150	Adults	Germany	150 adults ($M = 23.41$, $SD = 7.07$), 22.93 % males	Social Phobia Inventory (SPI)	Non-clinical	Executive Functioning	Attention Network Test	4
Cohen et al. (1996)	49	Adults	USA	SAD group: 17 adults ($M = 34.0$, $SD = 9.0$), 41.18 % males HC group: 32 adults ($M = 34.0$, $SD = 8.98$), 59.38 % males	Psychiatric screening for DSM-III-R diagnoses	Clinical	Cognitive Flexibility, Working Memory	Trail-making test (TMT), WAIS-R Digit Span	4
Demetriou et al. (2021)	136	Adults	Australia	SAD group: 82 adults ($M = 22.0$, $SD = 5.5$), 42.3 % males HC group: 54 adults ($M = 25.0$, $SD = 5.5$), 17.9 % males	Anxiety Diagnostic Interview Schedule (ADIS), Social Interaction Anxiety Scale (SIAS)	Clinical	Cognitive Flexibility, Executive Functioning	CANTAB-IED, TMT-B, BRIEF	3
Demetriou et al. (2018)	135	Adults	Australia	SAD group: 76 adults ($M = 22.11$, $SD = 5.64$), 41.18 % males HC group: 59 adults ($M = 24.88$, $SD = 5.3$), 59.38 % males	Standardised Clinical interviews, Social Interaction Anxiety Scale (SIAS)	Clinical	Cognitive Flexibility, Working Memory, Inhibitory Control, Executive Functioning	CANTAB-IED, TMT-B, BRIEF	3
Evans et al. (2018)	85	Adults	USA	85 adults ($M = 19.41$, $SD = 1.47$), 43.53 % males	Liebowitz Social Anxiety Scale (LSAS)	Non-clinical	Inhibitory Control	Flanker task	3
Filippi et al. (2022)	127	Children/Adolescents	USA	127 children ($M = 12$, $SD = 0$), 46.4 % males	Semi-structured diagnostic interviews, Screen for Child Anxiety Related Disorders	Clinical	Inhibitory Control	Flanker task	5

(continued on next page)

Table 1 (continued)

Study	N	Age group	Country	Sample Characteristics	Measure of SAD	Population	EF of Interest	EF Measure	MMAT Score
Fujii et al. (2013)	60	Adults	Japan	SAD group: 30 adults ($M = 23.9$, $SD = 6.7$), 66.67 % males HC group: 30 adults ($M = 25.6$, $SD = 5.6$), 66.67 % males	Mini International Neuropsychiatric Interview (MINI)	Clinical	Cognitive Flexibility, Executive Functioning	TMT-B, WCST	4
Gómez-Ariza et al. (2013)	164	Adolescents	Spain	SAD group: 60 adolescents ($M = 15.03$, $SD = 1.43$), 37 % males HC group: 104 adolescents ($M = 15.29$, $SD = 1.34$), 44 % males	Social Anxiety Scale for Adolescents (SAS-A), the Anxiety Disorders Interview Schedule for DSM-IV: Child Version	Clinical	Inhibitory Control	Novel task (selective list method-directed forgetting procedure)	4
Gorlin and Teachman (2015a)	159	Adults	USA	159 adults ($M = 18.7$, $SD = 1.19$), 29.3 % males	The Social Interaction Anxiety Scale (SIAS)	Non-clinical	Inhibitory Control	Stroop task	3
Gorlin and Teachman (2015b)	135	Adults	USA	135 adults ($M = 18.70$, $SD = 1.25$), 29.1 % males	Social Phobia Scale (SPS), Brief Fear or Negative Evaluative Scale, Social Interaction Anxiety Scale (SIAS)	Non-clinical	Inhibitory Control	Stroop task	3
Graver and White (2007)	33	Adults	USA	SAD group: 22 adults ($M = 21.80$, $SD = 3.43$), 9.09 % males HC group: 11 adults ($M = 21.11$, $SD = 7.49$), 27.27 % males	Social Interaction Anxiety Scale (SIAS), Anxiety Disorders Interview Schedule for DSM-IV Disorders	Clinical	Executive Functioning, Working Memory, Cognitive Flexibility	WCST, Digit Span task, TMT	3
Grisham and Williams (2013)	50	Adults	Australia	SAD group: 25 adults ($M = 29.44$, $SD = 13.78$), 32.0 % males HC group: 25 adults ($M = 33.84$, $SD = 9.38$), 16.0 % males	The Anxiety Disorders Interview Schedule for DSM-IV	Clinical	Cognitive Flexibility, Working Memory, Inhibitory Control	TMT-B, Operation-Span Task, the Hayling Sentence Completion Test	4
Heenan and Troje (2015)	48	Adults	Canada	48 adults ($M = 18.91$, $SD = 1.18$), 27.08 % males	Social Interaction Anxiety Scale (SIAS)	Non-clinical	Inhibitory Control	Go/No-Go task	3
Heeren et al. (2015)	50	Adults	Belgium	SAD group: 25 adults ($M = 47.28$, $SD = 12.93$), 24 % males HC group: 25 adults ($M = 48.08$, $SD = 12.78$), 24 % males	Mini International Neuropsychiatric Interview (MINI), Liebowitz Social Anxiety Scale (LSAS)	Clinical	Executive Functioning	Attention Network Test	4
Holder et al. (2020)	49	Adults	Australia	49 adults ($M = 23.12$, $SD = 6.37$), 22 % males	Social Interaction Anxiety Scale (SIAS), Social Phobia Scale (SPS)	Non-clinical	Executive Functioning	WCST percent pers errors	4
Judah et al. (2016)	42	Adults	USA	High SA group: 20 adults ($M = 19.80$, $SD = 5.34$), 45 % males Low SA group: 22 adults ($M = 18.95$, $SD = 1.81$), 32 % males	Social Interaction Anxiety Scale (SIAS)	Non-clinical	Cognitive Flexibility	Letter-number Sequencing task	3
Judah et al. (2013)	32	Adults	USA	32 adults ($M = 19.78$, $SD = 1.29$), 44 % males	Social Interaction Anxiety Scale (SIAS)	Non-clinical	Working Memory	Attentional Control Scale	3
Kraft and Grant (2021)	61	Adults	USA	61 adults ($M = 19.18$, $SD = 1.43$), 28 % males	Social Interaction Anxiety Scale (SIAS)	Non-clinical	Inhibitory Control	Go/No-Go task	3
Kraft, Grant, White, et al. (2021)	80	Adults	USA	80 adults ($M = 19.01$, $SD = 1.32$), 20 % males	Social Interaction Anxiety Scale (SIAS)	Non-clinical	Inhibitory Control	Attentional Control Scale	3
Liang (2018)	83	Adults	Taiwan	High SA group: 42 adults ($M = 20.79$, $SD = 1.73$), 33.33 % males Low SA group: 41 adults ($M = 20.90$, $SD = 1.32$), 21.95 % males	Brief Fear of Negative Evaluation Scale (BFNE), Social Interaction Anxiety Scale (SIAS)	Non-clinical	Cognitive Flexibility, Inhibitory Control	Mixed Antisaccade Task	3

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Table 1 (continued)

Study	N	Age group	Country	Sample Characteristics	Measure of SAD	Population	EF of Interest	EF Measure	MMAT Score
Lievore et al. (2025)	67	Children/Adolescents	Italy	67 children/adolescents ($M = 12.72$, $SD = 1.65$), 44.78 % males	Multidimensional Anxiety Scale for Children (MASC)	Non-clinical	Inhibitory Control	Go/No-Go Task	3
Mohlman and DeVito (2017)	80	Adults	USA	High SA group: 40 adults ($M = 23.03$, $SD = 7.01$), 50 % males Low SA group: 40 adults ($M = 22.31$, $SD = 4.98$), 45 % males	Social Phobia Scale (SPS), Fear of Negative Evaluation Scale (FNE)	Non-clinical	Executive Functioning	WCST	3
Moore et al. (2016)	50	Adults	Spain	SAD group: 25 adults ($M = 21.08$, $SD = 1.75$), 5 % males HC group: 25 adults ($M = 20.44$, $SD = 1.33$), 18 % males	Anxiety Diagnostic Interview Schedule (ADIS), Brief Social Phobia and Anxiety Inventory (SPAI-B), the Brief Fear of Negative Evaluation Scale (BFNE), Social Phobia Inventory (SPIN)	Clinical	Inhibitory Control	Auditory Proactive Interference Task	3
Morea and Calvete (2023)	680	Adolescents	Spain	680 adolescents ($M = 14.58$, $SD = 1.36$), 58.82 % males	Social Anxiety Scale for Adolescents (SAS-A)	Non-clinical	Executive Functioning	Perceived Executive Functions (PEF)	5
Morea and Calvete (2021)	698	Adolescents	Spain	698 adolescents ($M = 14.59$, $SD = 1.36$), 59.17 % males	Social Anxiety Scale for Adolescents (SAS-A)	Non-clinical	Cognitive Flexibility	Changes Cognitive Flexibility Test	4
Moriya (2024)	105	Adults	Japan	105 adults ($M = 38.40$, $SD = 7.78$), 46.67 % males	Brief Fear of Negative Evaluation Scale (BFNE), Social Phobia Scale (SPS), Social Interaction Anxiety Scale (SIAS)	Non-clinical	Inhibitory Control, Working Memory	Visual-search task	4
Moriya (2018)	117	Adults	Japan	117 adults ($M = 19.6$, $SD = 1.0$), 17.95 % males	Brief Fear of Negative Evaluation Scale (BFNE)	Non-clinical	Executive Functioning, Working Memory	Attention Network Test, Change Detection Task	3
Moriya and Sugiura (2013)	40	Adults	Japan	40 adults ($M = 19.5$, $SD = 2.0$), 45 % males	Brief Fear of Negative Evaluation Scale (BFNE)	Non-clinical	Inhibitory Control, Working Memory	Change detection task, Spatial-blink task	3
Moriya and Sugiura (2012)	50	Adults	Japan	50 adults (18–27 years), 38 % males	Brief Fear of Negative Evaluation Scale (BFNE), State Trait Anxiety Inventory-State form (STAI-T)	Non-clinical	Working Memory	Change detection task	3
Okawa et al. (2020)	72	Adults	Japan	SAD group: 20 adults ($M = 27.20$, $SD = 9.24$), 65 % males HC group: 52 adults ($M = 26.96$, $SD = 10.21$), 67.31 % males	Mini International Neuropsychiatric Interview for DSM-IV (MINI), Liebowitz Social Anxiety Scale (LSAS)	Clinical	Cognitive Flexibility	TMT-B	3
O'Toole et al. (2015)	84	Adults	Denmark	SAD group: 42 adults ($M = 29.20$, $SD = 7.90$), 36 % males HC group: 42 adults ($M = 25.40$, $SD = 6.60$), 36 % males	Anxiety Disorders Interview Schedule (ADIS-IV), Liebowitz Social Anxiety Scale (LSAS)	Clinical	Executive Functioning, Working Memory	Brixton Spatial Anticipation Test (BSAT), Digit Span, Letter-Number Sequencing Tests	4
Peschard et al. (2013)	36	Adults	Belgium	High SA group: 18 adults ($M = 20.10$, $SD = 1.90$), 50 % males Low SA group: 18 adults ($M = 19.89$, $SD = 1.2$), 50 % males	Liebowitz Social Anxiety Scale (LSAS)	Non-clinical	Inhibitory Control	Stroop task	3
Sachs et al. (2004)	50	Adults	Austria	SAD group: 25 adults ($M = 29.50$, $SD = 7.0$), 52 % males HC group: 25 adults ($M = 29.50$, $SD = 7.0$), 52 % males	Structured Clinical Interview for DSM-IV	Clinical	Executive Functioning	WCST	4

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Table 1 (continued)

Study	N	Age group	Country	Sample Characteristics	Measure of SAD	Population	EF of Interest	EF Measure	MMAT Score
Sutterby and Bedwell (2012)	50	Adults	USA	SAD group: 25 adults ($M = 38.04$, $SD = 12.85$), 52 % males HC group: 25 adults ($M = 38.60$, $SD = 12.10$), 52 % males	Anxiety Disorders Interview Schedule (ADIS-IV), Social Phobia and Anxiety Inventory (SPAI)	Clinical	Working Memory, Cognitive Flexibility, Inhibitory Control	Letter-number Sequencing task, Spatial Span, TMT, Stroop	4
Topçuoğlu et al. (2009)	72	Adults	Turkey	SAD group: 36 adults ($M = 26.20$, $SD = 7.15$), 55.6 % males HC group: 36 adults ($M = 26.30$, $SD = 4.7$), 55.6 % males	Structured Clinical Interview for DSM-IV Axis I Disorders - Clinical Version (SCID-I CV), Liebowitz Social Anxiety Scale (LSAS)	Clinical	Executive Functioning	WCST	4
Troller-Renfree et al. (2015)	55	Adults	USA	High SA group: 29 adults ($M = 19.15$, $SD = 1.29$), 46.2 % males Low SA group: 26 adults ($M = 18.87$, $SD = 1.01$), 47.8 % males	Liebowitz Social Anxiety Scale (LSAS)	Non-clinical	Executive Functioning, Inhibitory Control, Working Memory	Dimensional Change Card Sorting task, Flanker task, List Sorting Working Memory	3
Van Der Linden et al. (2005)	32	Adults	Switzerland	SAD group: 16 adults ($M = 36.69$, $SD = 7.33$), 43.75 % males HC: 16 adults ($M = 36.13$, $SD = 9.65$), 50 % males	Semi-structured interview for DSM-IV diagnosis, Liebowitz Social Anxiety Scale (LSAS)	Clinical	Inhibitory Control	Hayling task	3
Waechter et al. (2018)	52	Adults	Canada	SAD group: 25 adults ($M = 31.43$, $SD = 9.66$), 41 % males HC group: 27 adults ($M = 25.48$, $SD = 9.39$), 32 % males	Structured Clinical Interview for DSM-IV (SCID-I), Mini International Neuropsychiatric Interview (MINI), Social Phobia Inventory (SPIN)	Clinical	Working Memory	Operation Span task	4
Wang et al. (2020)	156	Adults	China	156 adults ($M = 20$), 35.90 % males	Interaction Anxiety Scale (IAS)	Non-clinical	Cognitive Flexibility, Executive Functioning	Cognitive Flexibility Scale (CFS), Attentional Control Scale (ACS)	3
Yuan et al. (2020)	34	Adults	China	High SA group: 17 adults ($M = 23.24$, $SD = 3.49$), 17.64 % males Low SA group: 17 adults ($M = 23.35$, $SD = 2.29$), 29.41 % males	Liebowitz Social Anxiety Scale (LSAS)	Non-clinical	Working Memory	Change Detection task	3
Zhao et al. (2020)	45	Adults	China	45 adults (N.S)	Liebowitz Social Anxiety Scale (LSAS)	Non-clinical	Inhibitory Control, Working Memory	Stroop task, Automated Operation Span task	3
Zhao et al. (2024)	42	Adults	China	High SA group: 21 adults ($M = 21.76$, $SD = 2.12$), 28.57 % males Low SA group: 21 adults ($M = 22.67$, $SD = 1.74$), 33.33 % males	Liebowitz Social Anxiety Scale (LSAS)	Non-clinical	Executive Functioning	AX-Continuous performance task	3

Note. N = number of participants, SAD = Social Anxiety Disorder, EF = Executive function, MMAT = Mixed Methods Appraisal Tool.

[-0.29, -0.10], $p < .001$). As there was no significant difference between the studies, the findings were considered robust to the assumption of independence of errors.

A sensitivity analysis was also completed for study design. The overall mean effect size was small and significant for group comparison studies ($r = -.22$, 95 % CI [-0.32, -0.11], $p < .001$), and was small and non-significant for correlational studies ($r = -.06$ 95 % CI [-0.16, 0.04], $p = .219$), indicating that study design was a factor that impacted the findings. The correlational studies typically measured social anxiety in a community sample, whilst the group comparison studies involved comparison between individuals with clinical levels of SAD and HCs,

which may have contributed to this discrepancy.

7. Meta-analysis of included studies

A total of 49 studies were found to examine the link between social anxiety and EF at different ages across the lifespan (see Fig. 2). The analysis revealed a small and significant effect size ($r = -.15$, 95 % CI [-0.22, -0.08], $p < .001$), with significant heterogeneity evident between the studies, $Q(48) = 462.74$, $p < .001$, $I^2 = 89.63$ %. No significant asymmetry was observed in the funnel plot, which was confirmed by Egger's test ($\beta = 1.33$, $SE = 1.06$, $p = .214$). However, trim-and-fill

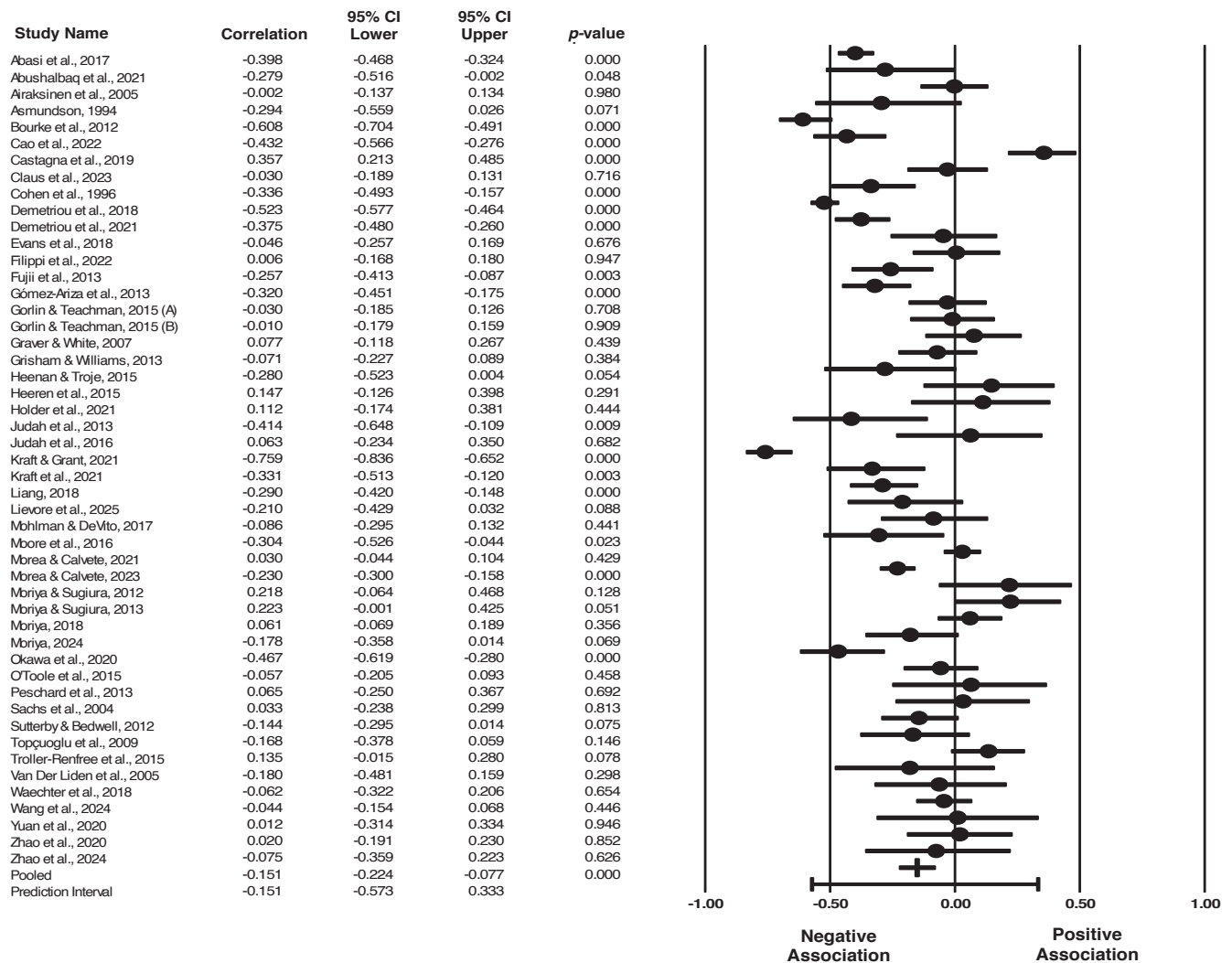


Fig. 2. Forest Plot Displaying the Effect Size Estimates for Studies Included in the Review.

Table 2

Summary and Moderator Effect Sizes for Link Between Social Anxiety and Executive Functioning Across Cognitive Domains.

Moderators / Categories	Cognitive Domains									
	Overall		Global Executive Functioning		Cognitive Flexibility		Inhibitory Control		Working Memory	
	r	k	r	k	r	k	r	k	r	k
Overall	-.151***	49	-.168*	17	-.204**	15	-.182**	19	-.058	18
Age Group										
Children/ Adolescents	-.060	6	-.230***	1	-.030	1	-.179*	3	-.357***	1
Adults	-.166***	43	-.161	16	-.225***	14	-.182*	16	-.086	17
SAD Severity										
Clinical	-.214***	21	-.245*	8	-.196**	10	-.225**	7	-.241**	9
Non-Clinical	-.104*	28	-.105*	9	-.222	5	-.157	12	.131	9
Assessment Type										
Cognitive Task	-.119**	44	-.054	12	-.186**	12	-.160*	18	-.017	17
Self-report	-.376***	7	-.417**	5	-.217	4	-.505***	1	-.607***	1
Task Index										
Reaction Time	-.204*	25	.057	3	-.225***	11	-.322	9	.077	2
Accuracy	-.065	45	-.065	11	-.009	5	-.119	14	-.029	15
Comorbidity										
No	-.263***	5	-.274*	1	-.260**	2	-.253**	2	-.045	1
Yes	-.139**	44	-.161*	16	-.197**	13	-.173*	17	-.279*	17

Note. * $p \leq .05$, ** $p \leq .01$, *** $p \leq .001$. k represents the number of studies included in analysis, r represents the effect size (.10 = small, .30 = moderate, .50 = large), SAD = Social Anxiety Disorder.

analysis identified potential publication bias (see Fig. 4), imputing 9 studies with negative effect sizes ($r_{\text{adjusted}} = -.22$, 95 % CI [-0.24, -0.20]). Nevertheless, the interpretation of the analysis remained unchanged, suggesting that the potential impact of publication bias was trivial.

8. Moderator analyses

Subgroup analyses were performed to explore the significant heterogeneity of effect sizes reported between the studies. Assessment type was the only significant moderator of EF performance, $Q(1) = 8.84$, $p = .003$, revealing that larger effects of SAD on EF performance were observed for self-report data than for cognitive tasks. However, both the self-report and cognitive task subgroups reached statistical significance. Age, $Q(1) = 1.06$, $p = .303$, SAD severity, $Q(1) = 2.28$, $p = .131$, task index, $Q(1) = 1.81$, $p = .179$, comorbidity, $Q(1) = 3.51$, $p = .061$, and EF domain, $Q(3) = 1.92$, $p = .589$, were not found to moderate the link between SAD and EF performance (see Table 2 for a summary of the moderator statistics).

9. Individual cognitive domains

The four cognitive domains examined across the studies were inhibitory control, cognitive flexibility, working memory, and global EF (see Fig. 3).

9.1. Inhibitory control

Nineteen studies used a measure of inhibitory control (e.g., Stroop task, Flanker task) to assess participants. The overall mean effect size was small and significant ($r = -.18$, 95 % CI [-0.31, -0.05], $p = .006$), and significant heterogeneity was reported across the studies, $Q(18) = 122.79$, $p < .001$, $I^2 = 85.34$ %. Whilst Egger's test was non-significant ($\beta = 2.62$, $SE = 2.31$, $p = .271$), four studies with negative effect sizes were imputed following trim-and-fill analysis, ($r_{\text{adjusted}} = .26$, 95 % CI [-0.30, -0.21]), indicating modest influence of publication bias. Assessment type was the only significant moderator of EF performance, $Q(1) = 14.09$, $p < .001$, with self-report data showing a larger, significant effect. All other moderators were non-significant (all $ps > .410$).

EF Domain	Study Name	Correlation	95% CI Lower	95% CI Upper	p-value
Cognitive Flexibility	Abasi et al., 2017	-0.398	-0.498	-0.294	0.000
Cognitive Flexibility	Aralainen et al., 2005	-0.002	-0.137	0.134	0.990
Cognitive Flexibility	Asmundson, 1994	-0.294	-0.559	0.026	0.071
Cognitive Flexibility	Cohen et al., 1996 (a)	-0.490	-0.663	-0.267	0.000
Cognitive Flexibility	Demetriou et al., 2018	-0.139	-0.298	0.027	0.100
Cognitive Flexibility	Demetriou et al., 2021	-0.119	-0.279	0.048	0.162
Cognitive Flexibility	Fuji et al., 2013 (a)	-0.280	-0.458	0.004	0.054
Cognitive Flexibility	Grauer & White, 2007 (a)	-0.163	-0.171	0.463	0.340
Cognitive Flexibility	Grisham & Williams, 2013 (a)	-0.152	-0.404	0.122	0.276
Cognitive Flexibility	Judith et al., 2013	-0.414	-0.648	-0.109	0.009
Cognitive Flexibility	Liang, 2018 (a)	-0.298	-0.476	-0.097	0.004
Cognitive Flexibility	Morea & Calvete, 2021	0.030	-0.044	0.104	0.429
Cognitive Flexibility	Okawa et al., 2020	-0.467	-0.619	-0.280	0.000
Cognitive Flexibility	Sutterby & Bedwell, 2012 (a)	-0.161	-0.412	0.113	0.248
Cognitive Flexibility	Wang et al., 2024 (a)	-0.048	-0.204	0.110	0.552
Cognitive Flexibility	Pooled	-0.204	-0.314	-0.089	0.001
Cognitive Flexibility	Prediction Interval	-0.204	-0.592	0.245	
Executive Functioning	Claus et al., 2023	-0.030	-0.189	0.131	0.716
Executive Functioning	Demetriou et al., 2018 (a)	-0.680	-0.752	-0.592	0.000
Executive Functioning	Demetriou et al., 2021 (a)	-0.658	-0.757	-0.529	0.000
Executive Functioning	Fuji et al., 2013	-0.274	-0.485	-0.032	0.027
Executive Functioning	Grauer & White, 2007	0.084	-0.251	0.400	0.630
Executive Functioning	Heeren et al., 2015	0.147	-0.126	0.398	0.291
Executive Functioning	Holder et al., 2021	0.112	-0.174	0.381	0.444
Executive Functioning	Kraft et al., 2021	-0.331	-0.513	-0.120	0.003
Executive Functioning	Mohman & DeVito, 2017	-0.086	-0.295	0.132	0.441
Executive Functioning	Morea & Calvete, 2023	-0.230	-0.300	-0.159	0.000
Executive Functioning	Monja, 2018 (a)	-0.070	-0.248	0.113	0.454
Executive Functioning	O'Toole et al., 2015 (a)	-0.108	-0.311	0.104	0.319
Executive Functioning	Sachs et al., 2004	0.033	-0.238	0.299	0.813
Executive Functioning	Toppooglu et al., 2009	-0.168	-0.378	0.059	0.146
Executive Functioning	Trollier-Renfroe et al., 2015	-0.021	-0.278	0.238	0.876
Executive Functioning	Wang et al., 2024	-0.039	-0.195	0.119	0.629
Executive Functioning	Zhao et al., 2024	-0.075	-0.369	0.223	0.626
Executive Functioning	Pooled	-0.188	-0.257	-0.034	0.015
Executive Functioning	Prediction Interval	-0.188	-0.634	0.257	
Inhibitory Control	Cao et al., 2022	-0.432	-0.566	-0.276	0.000
Inhibitory Control	Demetriou et al., 2018 (b)	-0.505	-0.613	-0.378	0.000
Inhibitory Control	Evans et al., 2018	-0.046	-0.257	0.169	0.676
Inhibitory Control	Filippi et al., 2022	0.006	-0.168	0.180	0.947
Inhibitory Control	Gómez-Ariza et al., 2013	-0.380	-0.451	-0.175	0.000
Inhibitory Control	Gorlin & Teichman, 2015 (A)	-0.030	-0.185	0.125	0.708
Inhibitory Control	Gorlin & Teichman, 2015 (B)	-0.010	-0.179	0.159	0.939
Inhibitory Control	Grisham & Williams, 2013 (b)	-0.065	-0.330	0.208	0.642
Inhibitory Control	Heenan & Troje, 2015	-0.280	-0.523	0.004	0.054
Inhibitory Control	Kraft & Grant, 2021	-0.759	-0.836	-0.692	0.000
Inhibitory Control	Liang, 2018	-0.281	-0.462	-0.078	0.007
Inhibitory Control	Lisakore et al., 2025	-0.210	-0.429	0.032	0.088
Inhibitory Control	Moore et al., 2016	-0.304	-0.526	-0.044	0.023
Inhibitory Control	Monja & Suglira, 2013	0.087	-0.230	0.388	0.594
Inhibitory Control	Peschard et al., 2013	0.065	-0.250	0.367	0.594
Inhibitory Control	Sutterby & Bedwell, 2012 (b)	-0.086	-0.348	0.188	0.539
Inhibitory Control	Trollier-Renfroe et al., 2015 (a)	0.221	-0.036	0.451	0.091
Inhibitory Control	Van Der Linden et al., 2005	-0.180	-0.481	0.159	0.298
Inhibitory Control	Zhao et al., 2020 (a)	0.140	-0.160	0.416	0.361
Inhibitory Control	Pooled	-0.182	-0.307	-0.052	0.006
Inhibitory Control	Prediction Interval	-0.182	-0.645	0.279	
Working Memory	Abushalab et al., 2021	-0.279	-0.516	-0.002	0.048
Working Memory	Bourie et al., 2012	-0.608	-0.704	-0.491	0.000
Working Memory	Castagna et al., 2019	0.357	0.213	0.485	0.000
Working Memory	Chen et al., 1998	-0.138	-0.336	0.138	0.326
Working Memory	Demetriou et al., 2018 (c)	-0.607	-0.684	-0.532	0.000
Working Memory	Grauer & White, 2007 (b)	-0.018	-0.344	0.312	0.919
Working Memory	Grisham & Williams, 2013	0.007	-0.264	0.277	0.959
Working Memory	Judith et al., 2016	0.063	-0.234	0.350	0.982
Working Memory	Monja & Suglira, 2012	0.218	-0.064	0.488	0.128
Working Memory	Monja & Suglira, 2013 (a)	0.350	0.043	0.596	0.026
Working Memory	Monja, 2018	0.100	0.009	0.359	0.040
Working Memory	Monja, 2024	-0.178	-0.369	0.014	0.069
Working Memory	O'Toole et al., 2015	-0.006	-0.215	0.204	0.968
Working Memory	Sutterby & Bedwell, 2012	-0.182	-0.428	0.060	0.189
Working Memory	Trollier-Renfroe et al., 2015 (b)	0.198	-0.061	0.432	0.132
Working Memory	Wlaichter et al., 2018	-0.062	-0.322	0.206	0.654
Working Memory	Yuan et al., 2020	0.012	-0.314	0.324	0.946
Working Memory	Zhao et al., 2020	-0.100	-0.382	0.189	0.516
Working Memory	Pooled	-0.058	-0.233	0.117	0.528
Working Memory	Prediction Interval	-0.058	-0.695	0.630	
Overall	Pooled	-0.169	-0.234	-0.101	0.000
Overall	Prediction Interval	-0.169	-0.601	0.341	

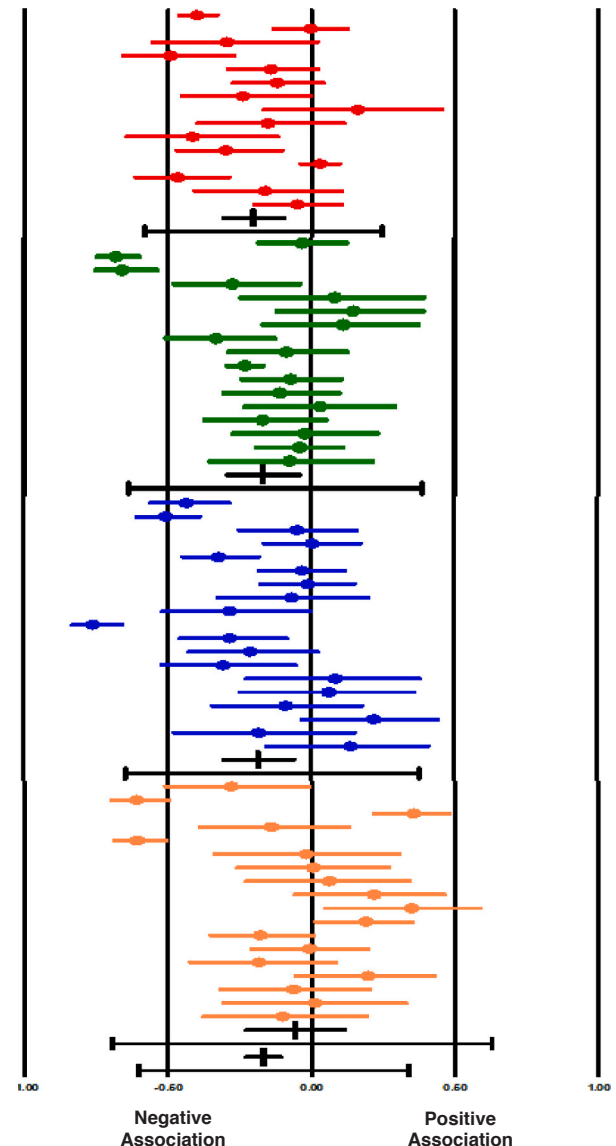


Fig. 3. Forest Plot Stratified by EF Domain.

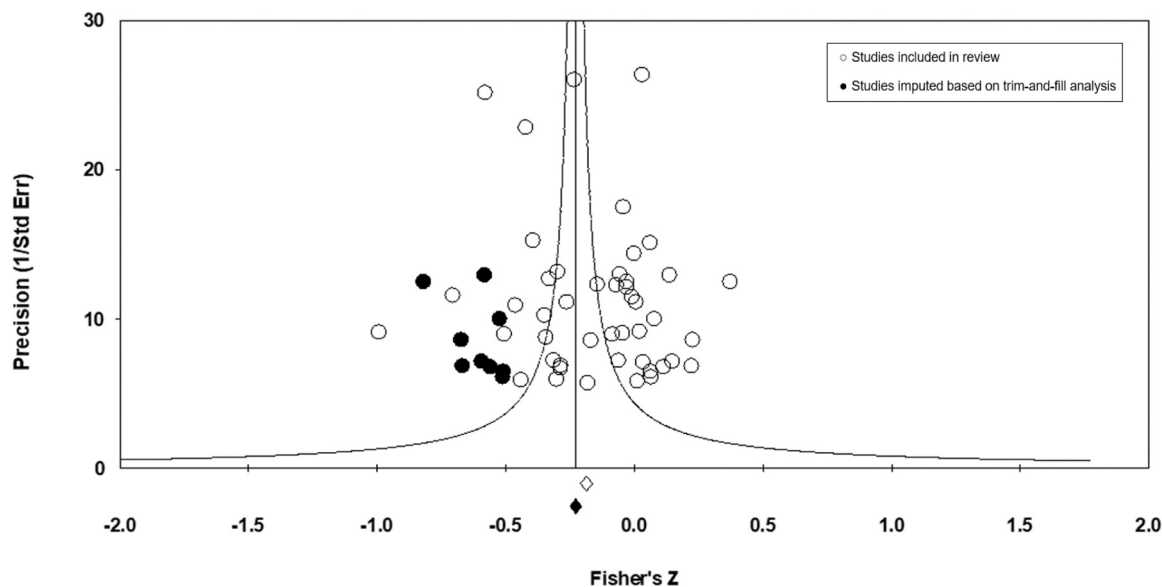


Fig. 4. Funnel Plot with Trim-and-Fill Studies Imputed.

9.2. Cognitive flexibility

Cognitive flexibility was assessed (e.g., Trail Making Test, Cognitive Flexibility Scale) in 15 studies involving participants with social anxiety. A small, significant effect size was observed ($r = -.20$, 95 % CI $[-0.31, -0.09]$, $p = .001$), with significant heterogeneity identified between the studies, $Q(14) = 94.98$, $p < .001$, $I^2 = 85.26\%$. The funnel plot and Egger's test indicated that there was no publication bias evident ($\beta = -1.17$, $SE = 1.43$, $p = .428$; no imputed studies). Age was a significant moderator in the analysis, $Q(1) = 14.47$, $p < .001$, with adults reporting a larger significant effect. In addition, task index was a significant moderator, $Q(1) = 10.13$, $p = .001$, with larger effects observed for RT as opposed to accuracy. All other moderators were non-significant (all p 's $> .579$).

9.3. Working memory

Eighteen studies examined working memory measures (e.g., Digit Span Task). The analysis revealed a small, non-significant effect size ($r = -.06$, 95 % CI $[-0.23, 0.12]$, $p = .528$). There was significant heterogeneity evident, $Q(17) = 177.98$, $p < .001$, $I^2 = 90.45\%$. The funnel plot and Egger's test was non-significant ($\beta = 3.76$, $SE = 2.78$, $p = .195$), although trim-and-fill analysis imputed six studies with negative effect sizes, to adjust for publication bias ($r_{\text{adjusted}} = -.26$, 95 % CI $[-0.30, -0.21]$). Moderator analyses revealed that age, $Q(1) = 14.90$, $p < .001$, SAD severity, $Q(1) = 7.67$, $p = .006$, and assessment type, $Q(1) = 36.68$, $p < .001$, were significant moderators, while task index, $Q(1) = 0.14$, $p = .712$, and comorbidity, $Q(1) = 1.93$, $p = .164$, were non-significant. A larger effect size was obtained for studies comprising youth compared to adults. Clinical samples showed a larger significant effect size than non-clinical samples, and self-report measures showed a larger significant effect size than cognitive tasks.

9.4. Global executive functioning

Measures of global EF (e.g., Wisconsin Card Sorting Test, BRIEF Global Executive Composite Score) were used in 17 studies throughout the literature. The effect size across the studies was small and significant ($r = -.17$, 95 % CI $[-0.30, -0.03]$, $p = .015$) and there was significant heterogeneity reported, $Q(16) = 134.55$, $p < .001$, $I^2 = 88.12\%$. There was no significant asymmetry evident in the funnel plot, which was supported by Egger's test ($\beta = 1.77$, $SE = 1.68$, $p = .310$). However, five

studies were imputed following trim-and-fill analysis, ($r_{\text{adjusted}} = -.26$, 95 % CI $[-0.30, -0.22]$), which did not substantially alter the conclusions of the meta-analysis, indicating that it was robust to the influence of publication bias. Moderator analyses revealed that assessment type was the only significant moderator, $Q(1) = 6.30$, $p = .012$, with self-report data showing a larger, significant mean effect size. All other moderators were non-significant (all p 's > 0.347).

9.5. Age group differences in EF

Although age was not a significant moderator in the analysis, differences in EF performance were evident between adults and youth. There was a small and significant effect for cognitive flexibility ($r = -.225$, 95 % CI $[-0.327, -0.117]$, $p < .001$), and inhibitory control ($r = -.182$, 95 % CI $[-0.330, -0.025]$, $p = .023$), for adults, but not for children (both p 's $> .05$). In contrast, a small and significant effect was reported for social anxiety and overall EF ($r = -.230$, 95 % CI $[-0.300, -0.158]$, $p < .001$), as well as working memory ($r = -.357$, 95 % CI $[0.213, 0.485]$, $p < .001$), for children, and not for adults (both p 's $> .05$), suggesting that the relationship between SAD and EF may differ across the lifespan (see Table 2 for a summary of findings).

10. Discussion

Studies that have examined the link between SAD and EF have produced mixed results to date, making it difficult to draw definitive conclusions regarding this relationship. The current systematic review and meta-analysis aimed to provide a comprehensive examination of the link between SAD and EF across the lifespan, and examine whether several important moderators could explain the heterogeneity reported in the literature. The findings revealed a significant relationship between SAD and EF, such that individuals with social anxiety performed significantly worse overall, on measures of EF than individuals with low or no social anxiety. These results are consistent with previous studies reporting that social anxiety is associated with deficits in EF, which may be a result of the reduced cortical thickness in the frontal brain regions that are associated with EF (Fujii et al., 2013; Graver & White, 2007; Rosa-Alcázar et al., 2021). However, the present review also suggests that, in certain contexts, assessment type, task index, symptom severity, and age can play a role in the relationship between SAD and EFs.

11. Moderating factors

Assessment type significantly moderated the relationship between SAD and EF, with stronger correlations evident for self-report data than for cognitive task performance. The stronger association for self-report data may be a result of the internalising symptoms associated with social anxiety. It is well documented that cognitive distortion and maladaptive core beliefs are a fundamental part of the symptomology of social anxiety. As a result, individuals with SAD are more likely to self-report greater deficits than may be indicated by cognitive performance. This has been shown throughout the literature, with studies reporting that social anxiety is linked to poorer self-reported EF performance (Demetriou et al., 2018; Judah et al., 2013; Kraft, Grant, White, et al., 2021; Morea & Calvete, 2023), which may be a result of the detrimental effect of these maladaptive cognitions causing individuals with social anxiety to report overly negative evaluations of their own functioning (Demetriou et al., 2018). This finding offers valuable insight into the relationship between SAD and EF, and suggests that self-report measures of EF may be influenced by biases in this clinical group.

Task index was not a significant moderator in the analysis. However, the correlations differed across measures, with a small, significant effect size reported for RT, while accuracy was non-significant. This suggests that individuals with social anxiety reacted significantly slower than healthy individuals, but demonstrated good accuracy, which is consistent with the attentional control theory (ACT). Rather than impairing performance effectiveness directly, ACT proposes that anxiety is associated with deficits in processing efficiency, leading to poorer performance as indicated by normal accuracy but slower response times during EF tasks (Derakshan & Eysenck, 2009; Eysenck & Derakshan, 2011; Eysenck et al., 2007). The slowing of RTs suggest that highly anxious individuals use alternative strategies, such as enhanced effort or greater use of processing resources, to compensate for anxiety induced cognitive deficits (Derakshan & Eysenck, 2009). As a result, performance effectiveness is typically maintained at the cost of performance efficiency or speed (Eysenck & Derakshan, 2011; Eysenck et al., 2007). Similar findings have been reported throughout the literature, with studies showing that SAD is associated with intact accuracy and poorer RT on measures of EF (Gayete et al., 2020; Majeed et al., 2023; Wong et al., 2013). This is more commonly reported for cognitive domains such as inhibitory control (Derakshan & Eysenck, 2009; Pacheco-Unguetti et al., 2010; Visu-Petra et al., 2013) and cognitive flexibility (Edwards et al., 2015; Visu-Petra et al., 2013) as these measures are particularly susceptible to the effects of anxiety due to their roles in attentional control, whereas the effects of anxiety on working memory are assumed to be weaker (Eysenck et al., 2007; Miyake et al., 2000b). This is consistent with the findings of the current study.

SAD severity also did not appear to moderate the relationship between SAD and EF. Despite this finding, the results revealed that in clinical samples, SAD was associated with significantly poorer EF, but this relationship was not significant in non-clinical samples. This difference is consistent with previous research (Bourke et al., 2012; Demetriou et al., 2018; Gómez-Ariza et al., 2013) that shows that clinical samples typically report greater effects, as they represent the extreme end of social anxiety, whilst the distribution for non-clinical samples typically clusters towards the lower end of the scale (Turner et al., 1989).

Finally, the results revealed that age was not a significant moderator for the relationship between SAD and EF. While these findings contradict previous research (Huizinga et al., 2006; Schmidt et al., 2022; Shi et al., 2019; Wang, 2022), this could be due to the discrepancy in the number of studies that examined youth compared to adults. The current meta-analysis included 6 studies that examined children and adolescent samples, whilst 43 studies examined adults. This could have resulted in lack of power to detect true statistical significance. Furthermore, the data were also more variable with the children and adolescent samples, indicating more error variance associated with the effects, which could

have contributed to these findings. Overall, the majority of the moderators included in this analysis were non-significant, which offers some support for the robustness of the relationship between SAD and EF performance. Individual differences such as gender and culture may be other important influences on the link between SAD and EF; however, insufficient data was reported in the studies for these variables, with only one study examining their potential impact on this relationship. This represents an important gap in the literature and therefore, future research is needed to better understand these effects.

12. Subdomains of EF

Differences in performance between individuals with social anxiety and controls were also reported for the subdomains of EF. Social anxiety was significantly associated with poorer inhibitory control, cognitive flexibility, and global EF performance, but no significant association was reported for working memory. This contradicts the findings of Abushalbak et al. (2021), Bourke et al. (2012), and Demetriou et al. (2018) who found that individuals with SAD demonstrated impaired working memory performance. However, several factors were shown to moderate this relationship, which may help to explain the non-significant findings reported in the current study. Age, symptom severity, and assessment type were all significant moderators in the relationship between SAD and working memory. The majority of the cognitive working memory tasks had no relationship with social anxiety (Graver & White, 2007; O'Toole & Pedersen, 2011; Yuan et al., 2020), while the self-report measure was strongly correlated (Demetriou et al., 2018). This suggests that SAD may particularly impact people's perception of their own working memory ability which may be a result of the maladaptive cognitions discussed previously. In addition, working memory was only significantly associated with social anxiety in youth which could indicate developmental differences that should be explored. Therefore, the association between SAD and working memory may be more nuanced than with other EF domains and warrants further investigation, specifically in regards to the impact of moderator variables.

13. Age group differences

Whilst age did not moderate the relationship between SAD and EF, it was evident that there were differences in performance across the individual domains of EF for youth and adults. Specifically, a significant relationship between social anxiety and measures of working memory and global EF was evident for children and adolescents, while studies with adults reported a significant relationship between social anxiety and cognitive flexibility as well as inhibitory control. The different patterns for the EF domains across the lifespan is consistent with research indicating that there are distinct developmental trajectories for each of the three core EFs (Conklin et al., 2007; Magar et al., 2010). This offers an area for additional research, as studies are yet to examine the link between SAD and EF specifically focusing on the developmental trajectory of these three core EFs.

14. Limitations, implications, and future research directions

Some limitations were evident in the studies that were reviewed. The current meta-analysis appeared to be robust to the influence of publication bias, and sensitivity analysis revealed that methodological quality did not significantly impact on the findings. In addition, independence of errors was not assumed for the analyses that included studies with multiple effect sizes, which has implications for effect size calculations and meta-analytic results. However, upon running the analysis (see [supplementary materials](#) for sensitivity analysis), there was no significant difference in the findings with and without these studies included. As the overall pattern of results remained consistent, the analysis is likely robust to the influence of nesting effects, further strengthening the credibility of results. Study design was found to influence the results of

the analysis, with correlational studies reporting non-significant findings overall, while group-comparison studies reported a small and significant overall mean effect size, potentially contributing to the high heterogeneity reported across studies. This discrepancy may have been a result of the different samples used, with correlational studies typically measuring social anxiety in subclinical, community samples, whilst the group comparison studies compared individuals with clinical levels of SAD and healthy controls. However, examining the link between social anxiety and EF in both clinical and subclinical populations is important for future research to provide a more representative and comprehensive examination of this relationship, especially given that social anxiety severity is thought to exist on a continuum, with varying symptoms and levels of impairment (Schneier et al., 2002; Stopa & Clark, 2001). To address this limitation, efforts should be made to ensure that community samples include individuals at the extreme ends of social anxiety, including individuals with clinical diagnoses.

In addition, the majority of the studies used single evaluation measures for EF, which does not offer a broad examination of the link between SAD and EF, and fails to consider the multidimensionality of this construct. There was also significant heterogeneity evident between the studies, which may have been a result of variability in study design and sample characteristics, as previously discussed. However, subgroup analyses also revealed that differences in the EF assessment methods impacted the outcomes (see [supplementary materials](#) for analysis). Specifically, a variety of tasks were used across studies, some that place high demands on cognitive processes other than EF, such as attention and processing speed, or measure different facets of the core EFs (i.e., flanker tasks measures interference control, whereas the Go/No-Go task measures response inhibition, two facets of inhibitory control). Therefore, the tasks may not yield a pure measure of the target EF, limiting comparability. Furthermore, there was a lack of studies using both self-report and behavioural assessments of EF. As the current review suggested that the effects were stronger for self-reports as opposed to behavioural measures, it is important that future research verifies and explores the source of this difference, which would help to provide a more detailed view of the relationship between SAD and EF. These inconsistencies could be contributing to the mixed findings reported in the literature. To address these limitations, future research is needed which employs a multi-method assessment approach, using several robust and pure measures of the three core EFs, including both self-reports and behavioural tasks, to provide a more comprehensive examination of the link between SAD and EF.

This study revealed that there was relatively little research in children and adolescent samples. Of the 49 studies included in this review, only 6 examined the link between SAD and EF in children and adolescents. The paucity of research in these earlier age groups is of concern given that these developmental stages represent the typical age of onset for SAD symptoms and EF development (Elizabeth et al., 2006; Theodoraki et al., 2020; Vogel et al., 2023), and therefore, may act as an important period for intervention. Future research should focus on examining the link between SAD and EF in all age groups, including children and adolescents, to better understand this relationship across the lifespan.

The direction of the relationship between SAD and EF across the lifespan remains inconclusive, with the scar and vulnerability models, two prominent theories in this body of literature, offering competing ideas on causality. Scar theories suggest that elevated anxiety and related symptoms predict a subsequent decline in EF performance whereas the vulnerability models in this context suggest that deficits in EF lead to the development and maintenance of anxiety disorders (Zainal & Newman, 2022). As the causal mechanisms underlying this relationship are unclear, future longitudinal research should be conducted to examine the relationship between SAD and EF across developmental periods, taking into consideration both the scar and vulnerability models. This could also offer explanatory value across diverse developmental stages by providing insight into the preceding

and predicting factors associated with the development of anxiety and EF deficits across the lifespan. In addition, while previous neuroimaging studies have examined neurological differences in individuals with SAD compared to HCs, research using longitudinal neuroimaging designs would further elucidate the causal relationship between SAD and EF by documenting the links with structural and functional changes across the lifespan. This has practical implications for understanding the development of EF deficits that may be evident in SAD and could assist with improving current intervention strategies.

The present review and meta-analysis has led to insights regarding our understanding of the psychopathology of SAD and the factors contributing to its development and maintenance. Importantly, this review identified that SAD is associated with deficits in EF, which supports the use of strategies to target this area in therapy. Specifically, research suggests that cognitive remediation training (CRT), which targets specific cognitive process through the use of repetitive cognitive exercises and training protocols (Wykes et al., 2018), can improve EF performance (Koster et al., 2017), with preliminary research supporting the efficacy of this treatment approach for individuals with SAD (Caudle et al., 2023; Hadwin & Richards, 2016; Pan et al., 2020, 2022). This approach may also maximise overall treatment efficacy, as EF deficits have been identified as barriers to therapy engagement and effectiveness, with studies showing that persisting cognitive impairments can reduce medication adherence and client retention (Dickson et al., 2023; Godovich et al., 2020; Gold et al., 2016). Therefore, new approaches that use cognitive remediation of EF adjunctive to psychological interventions could be developed to optimise treatment outcomes and improve both cognitive and overall functioning in individuals with social anxiety. Furthermore, examining the link between SAD and EF at all ages could provide insight into whether there are domain-specific deficits in EF that are evident at different stages across the lifespan. This information could allow for a more targeted and evidence-based approach to the selection of cognitive remediation training tasks in each age group. Depending on the age that functional impairments are observed, this could also provide insight into the key periods for prevention of lasting cognitive deficits.

15. Conclusion

Overall, this systematic review and meta-analysis provided a comprehensive review of existing research that has examined the link between SAD and EF, revealing that EF deficits are evident for individuals with social anxiety. Specifically, this clinically significant condition is associated with poorer performance on cognitive flexibility, inhibitory control, and global EF measures. While age did not significantly moderate the relationship between SAD and EF, there were differences in performance on specific EF domains for adults compared to children and adolescents, suggesting that the development of EFs may differ across the lifespan for individuals with social anxiety. Therefore, future research should aim to examine the development of the three core EFs in individuals with social anxiety compared to HCs and account for the potential impact of important moderators on this relationship. Continued research in this area is critical to help inform new strategies to improve daily functioning in this clinical population.

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CRediT authorship contribution statement

Karen Murphy: Writing – review & editing, Supervision, Methodology, Conceptualization. **David Neumann:** Writing – review & editing, Supervision, Conceptualization. **Alexandra Harrison:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal

analysis, Data curation, Conceptualization. **Lan Nguyen:** Writing – review & editing, Supervision, Methodology, Conceptualization, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

Subject Headings for Databases

Medical Subject Headings (MeSH) were used for PubMed, and standardised search terms were used for Ovid (PsychINFO) database searches. Scopus and ProQuest Health and Medical/ProQuest Psychology searches map to subject headings by default. The following are the exact search terms used for each database:

PubMed: ("youth" OR "young person*" OR "teen" OR "early adulthood" OR "child*" OR "adolescen*" OR "preadolescen*" OR "lifespan" OR "old* adult*" OR "adult*" OR "develop") OR ("adolescent" [MeSH] OR "adult" [MeSH] OR "aged" [MeSH] OR "child" [MeSH] OR "middle aged" [MeSH]) AND ("SAD" OR "social anxiet*" OR "social phobia" OR "social anxiety disorder*") OR ("phobia, social" [MeSH]) AND ("executive function*" OR "EF" OR "executive performance" OR "central executive" OR "executive control" OR "cognitive performance" OR "neuropsychological function" OR "neuropsychological test" OR "executive dysfunction" OR "cognitive flexibility" OR "shifting" OR "task switch*" OR "mental flexibility" OR "working memory" OR "WMC" OR "updating" OR "inhibition" OR "inhibitory control") OR ("executive function" [MeSH] OR "cognitive dysfunction" [MeSH] OR "inhibition, psychological" [MeSH] OR "memory, short-term" [MeSH]).

Ovid (PsychINFO): ("youth" OR "young person*" OR "teen" OR "early adulthood" OR "child*" OR "adolescen*" OR "preadolescen*" OR "lifespan" OR "old* adult*" OR "adult*" OR "develop") OR (exp older adulthood/ or exp early adolescence/ or exp emerging adulthood/ or exp aging/ or exp adult development/ or exp late adolescence/ or exp middle adulthood/ or exp adolescent development/ or exp lifespan/) AND ("SAD" OR "social anxiet*" OR "social phobia" OR "social anxiety disorder*") OR (exp social anxiety/ or exp social phobia/) AND ("executive function*" OR "EF" OR "executive performance" OR "central executive" OR "executive control" OR "cognitive performance" OR "neuropsychological function" OR "neuropsychological test" OR "executive dysfunction" OR "cognitive flexibility" OR "shifting" OR "task switch*" OR "mental flexibility" OR "working memory" OR "WMC" OR "updating" OR "inhibition" OR "inhibitory control") OR (exp executive function/ or exp executive functioning measures/ or exp cognitive flexibility/ or exp short term memory/ or exp response inhibition/ or exp cognitive control/).

Scopus: ("youth" OR "young person*" OR "teen" OR "early adulthood" OR "child*" OR "adolescen*" OR "preadolescen*" OR "lifespan" OR "old* adult*" OR "adult*" OR "develop") AND ("SAD" OR "social anxiet*" OR "social phobia" OR "social anxiety disorder*") AND ("executive function*" OR "EF" OR "executive performance" OR "central executive" OR "executive control" OR "cognitive performance" OR "neuropsychological function" OR "neuropsychological test" OR "executive dysfunction" OR "cognitive flexibility" OR "shifting" OR "task switch*" OR "mental flexibility" OR "working memory" OR "WMC" OR "updating" OR "inhibition" OR "inhibitory control").

ProQuest Health and Medical/ ProQuest Psychology: ("youth" OR "young person*" OR "teen" OR "early adulthood" OR "child*" OR "adolescen*" OR "preadolescen*" OR "lifespan" OR "old* adult*" OR "adult*" OR "develop") AND ("SAD" OR "social anxiet*" OR "social phobia" OR "social anxiety disorder*") AND ("executive function*" OR "EF" OR "executive performance" OR "central executive" OR "executive control" OR "cognitive performance" OR "neuropsychological function"

OR "neuropsychological test" OR "executive dysfunction" OR "cognitive flexibility" OR "shifting" OR "task switch*" OR "mental flexibility" OR "working memory" OR "WMC" OR "updating" OR "inhibition" OR "inhibitory control").

Appendix B

References for Articles Solely Included in the Meta-Analysis:

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Appendix C. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.janxdis.2025.103056](https://doi.org/10.1016/j.janxdis.2025.103056).

Data availability

This systematic review and meta-analysis was pre-registered on Open Science Framework. Data used in this study is freely available on the Open Science Framework repository: https://osf.io/ct8u7/?view_only=67f57782fce043fea2d837c5687d35c8

SAD - EF Review Dataset (Open Science Framework Repository)

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